

Day 6 Report — AI Internship

Topic: PCA for Dimensionality Reduction & Customer Segmentation using KMeans

Name: Shafiq ur Rehman

Day: 6

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Tech Stack: Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn

Objective

The objective of Day 6 was to:

- Apply **Principal Component Analysis (PCA)** for dimensionality reduction
 - Perform **customer segmentation using KMeans clustering**
 - Determine optimal number of clusters using **Elbow Method**
 - Validate clustering performance using **Silhouette Score**
 - Visualize customer segments using **PCA-based 2D plots**
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Dataset Overview

The dataset used was **Mall Customer Segmentation dataset**, which includes:

- CustomerID
 - Gender
 - Age
 - Annual Income (k\$)
 - Spending Score (1–100)
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Data Preprocessing

Before applying machine learning algorithms, the dataset was cleaned and prepared:

- Removed unnecessary column: **CustomerID**
- Handled missing values by removing incomplete rows
- Encoded categorical variable:
 - Gender → Male: 0, Female: 1
- Applied **StandardScaler** to normalize features

✓ This ensured all features were on the same scale for PCA and KMeans.



Principal Component Analysis (PCA)

PCA was applied to reduce dimensionality while preserving maximum variance in the data.

Steps performed:

- Converted original features into principal components
- Generated **explained variance ratio**
- Plotted **Scree Plot (Cumulative Explained Variance)**

Key Insight:

- Most of the variance was captured in the first two components
- This allowed reduction from 3D feature space to 2D

✓ PCA improved visualization and reduced computational complexity.



KMeans Clustering

KMeans algorithm was used to segment customers into groups.

Steps performed:

1. Elbow Method

- Tested K values from 2 to 8
- Plotted inertia (sum of squared distances)
- Identified optimal cluster count based on elbow point

2. Silhouette Score

- Evaluated clustering quality
 - Measured how well-separated clusters are
 - Selected best K based on highest score
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Final Model

- Applied KMeans with optimal number of clusters
- Assigned cluster labels to each customer

Each customer was grouped into a segment based on behavioral similarity.



Visualization

1. PCA Cluster Plot

- 2D visualization of customer segments
- Each cluster represented by a different color
- Clearly visible separation between groups

2. Cluster Interpretation (Observations)

- High income + high spending → VIP customers
 - Low income + high spending → impulsive buyers
 - Low spending score → inactive/low engagement customers
 - Moderate group → average customers
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Key Learnings

- PCA helps reduce dimensionality while preserving important variance
 - Scaling is essential before PCA and clustering
 - Elbow method helps estimate optimal number of clusters
 - Silhouette score validates clustering quality
 - KMeans is effective for customer segmentation problems
 - Visualization helps interpret ML results in business terms
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Conclusion

Day 6 focused on combining **dimensionality reduction (PCA)** with **unsupervised learning (KMeans clustering)** to segment customers effectively.

The model successfully grouped customers into meaningful clusters based on their spending behavior and income patterns, which can be used for targeted marketing strategies.